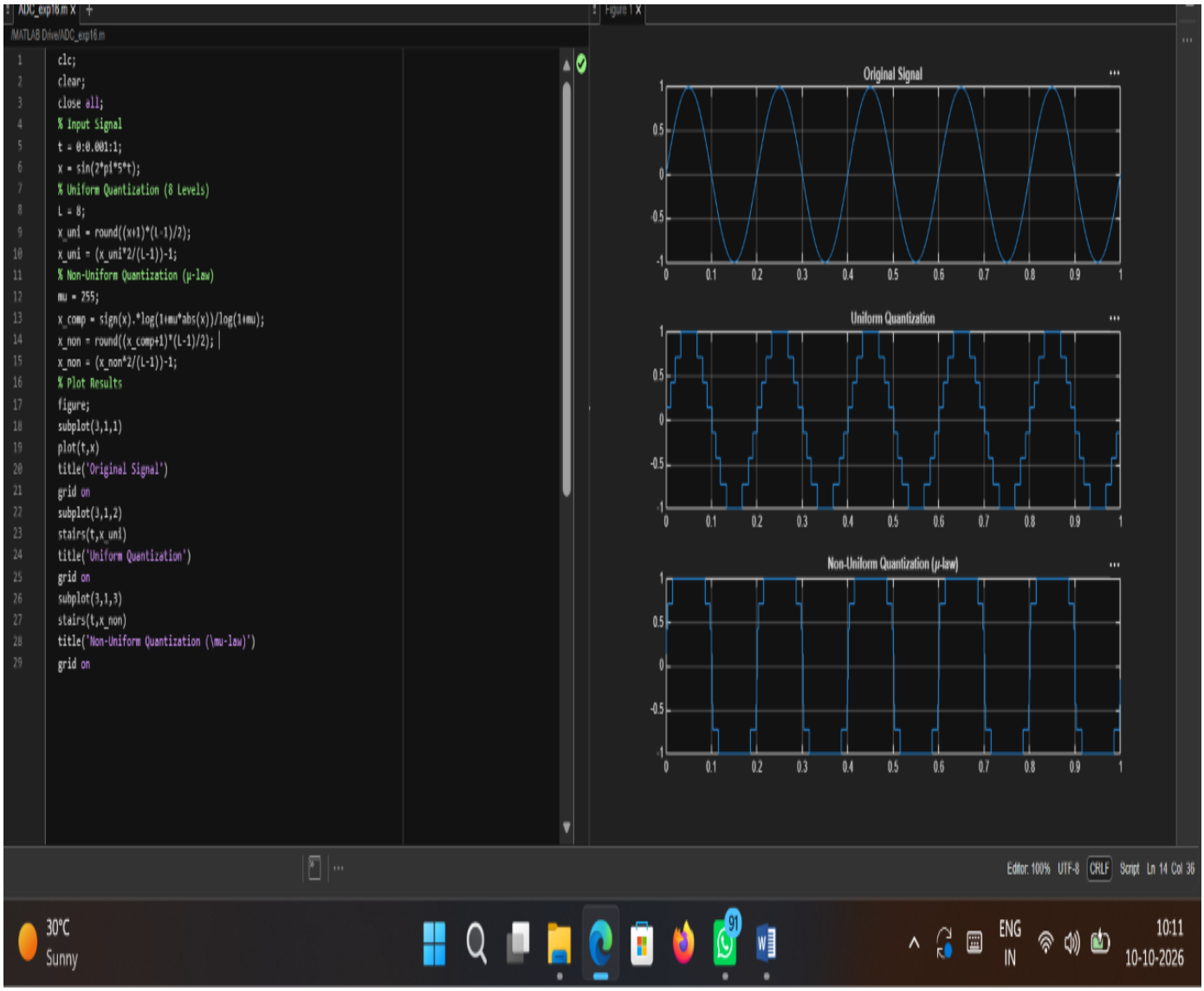
Ex.No:16: SIMULATION AND ANALYSIS OF UNIFORM AND NON-UNIFORM QUANTIZATION USING MATLAB

Objective 1:



Code:

```
clc;

clear;

close all;

% Input Signal

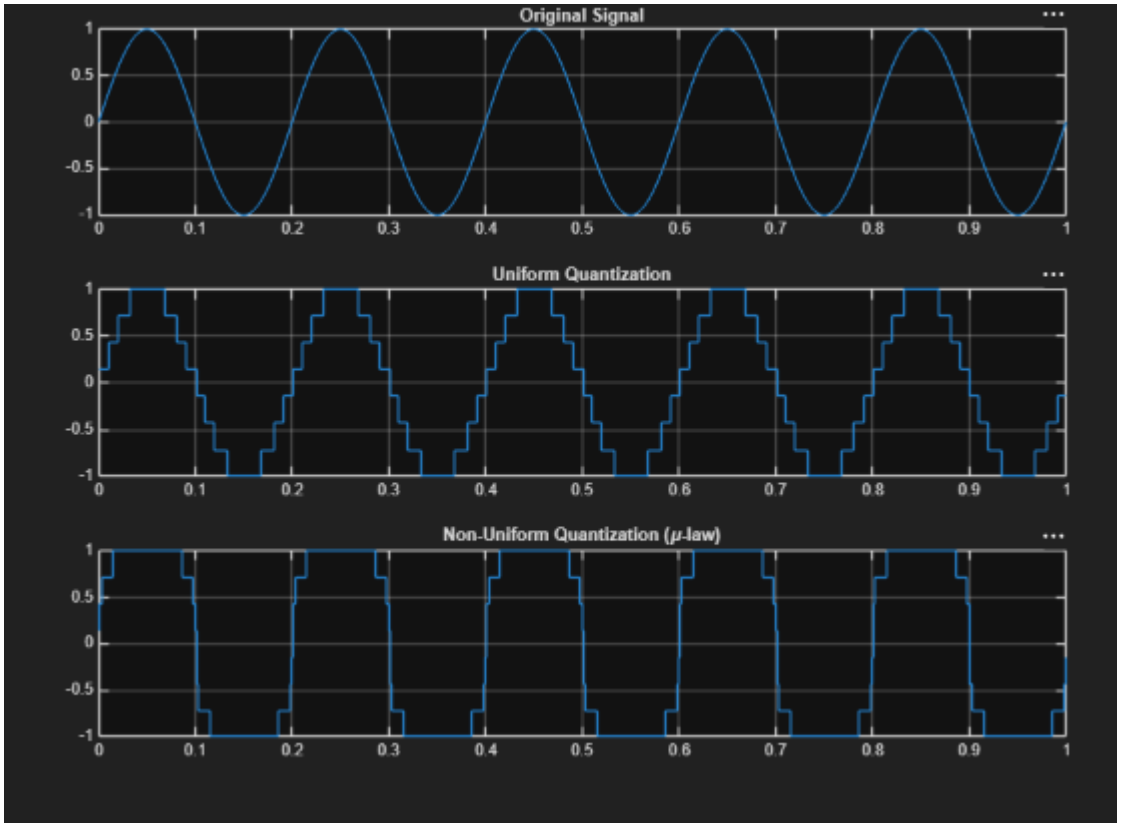
t = 0:0.001:1;

x = sin(2*pi*5*t);

% Uniform Quantization (8 Levels)
```

```
L = 8;
x_uni = round((x+1)*(L-1)/2);
x_uni = (x_uni*2/(L-1))-1;
% Non-Uniform Quantization (μ-law)
mu = 255;
x_comp = sign(x).*log(1+mu*abs(x))/log(1+mu);
x_non = round((x_comp+1)*(L-1)/2);
x_non = (x_non*2/(L-1))-1;
% Plot Results
figure;
subplot(3,1,1)
plot(t,x)
title('Original Signal')
grid on
subplot(3,1,2)
stairs(t,x_uni)
title('Uniform Quantization')
grid on
subplot(3,1,3)
stairs(t,x_non)
title('Non-Uniform Quantization (\mu-law)')
grid on
```

Output:


Original Signal
Uniform Quantization
Non-Uniform Quantization (μ-law)
1
0.5
0
-0.5
-1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1